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ON THE CONTRACTIVITY OF THE RIESZ PROJECTION

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Abstract. This note presents a proof that the Riesz projection is contractive from L^q to $L^{4(1-1/q)}$ for the case $1 < q < \infty$. This fills in the previously unresolved parameter range of [7, Conjecture 4].

1. The main result

Let $\mathbb{T}$ be the unit circle in the complex plane, equipped with the normalized Lebesgue measure dm. For $0 < p \leq \infty$, we denote by L^p the usual Lebesgue space on $\mathbb{T}$. It is well known that the Riesz projection (initially defined on trigonometric polynomials)

$$\mathcal{P}\left(\sum_{n\in\mathbb{Z}} c_n z^n\right) = \sum_{n\geq 0} c_n z^n$$

extends to a bounded operator on L^p if and only if $1 < p < \infty$. Moreover, the explicit norm

$$\|\mathcal{P}\|_{L^p\to H^p} = \csc(\pi/p), \quad 1 < p < \infty, \tag{1}$$

was determined by Hollenbeck and Verbitsky [11].

In this note, we consider the L^q-L^p contractivity of the Riesz projection: *for which values of q and p does $\mathcal{P}$ extend to a contraction from L^q to L^p?* This problem was first investigated by Marzo and Seip [15] and has since given rise to several interesting developments and applications. As pointed out in [15], by Hölder's inequality, this contractivity problem essentially amounts to determining the following *critical exponent of the Riesz projection*:

$$\mathfrak{p}(q) := \max\left\{p > 0 : \mathcal{P} : L^q \to L^p \text{ is contractive}\right\}.$$

By orthogonality, one immediately has $\mathfrak{p}(2) = 2$. Moreover, Marzo and Seip [15] proved that $\mathfrak{p}(\infty) = 4$ and hence obtained $\mathfrak{p}(4/3) = 1$ from the duality. Later, the upper bound

$$\mathfrak{p}(q) \leq 4\left(1 - \frac{1}{q}\right), \qquad 1 < q \leq \infty$$

was established by Brevig, Ortega-Cerdà, Seip, and Zhao in [7]. Motivated by a duality relation between certain contractive inequalities for Hardy spaces,

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Brevig, Ortega-Cerdà, Seip, and Zhao [7, Conjecture 4] then conjectured that

$$\mathfrak{p}(q) = 4\left(1 - \frac{1}{q}\right), \qquad 1 < q \leq \infty. \tag{2}$$

They also observed that Conjecture (2) predicts that the critical exponent tends to zero as q decreases to one. This naturally led them to conjecture the endpoint inequality

$$\|\mathcal{P}f\|_{L^0} := \exp\left(\int_{\mathbb{T}} \log|\mathcal{P}f|\, dm\right) \leq \|f\|_{L^1}. \tag{3}$$

Recently, Brevig, Llinares, and Seip [5] confirmed that the endpoint inequality (3) indeed holds for all $f \in L^1$. They also formulated a conjecture for the critical exponents of Riesz projections in several variables and proved that, if the one-dimensional critical exponent conjecture (2) is true, then the corresponding conjecture in two dimensions is also true [5, Corollary 4]. However, this implication does not necessarily extend to higher dimensions. We mention that these contractive inequalities have important applications to Hardy spaces on higher-dimensional tori and, via the Bohr lift, to Hardy spaces of Dirichlet series; see, for instance, [2, 3, 4, 6, 9, 12, 13, 14, 15].

The main result of this note is to prove that (2) is valid for all the remaining values $1 < q < \infty$.

Theorem 1.1. *For every $1 < q < \infty$, the Riesz projection*

$$\mathcal{P} : L^q \to L^{4(1-1/q)}$$

is contractive.

Our approach goes back to Cole's work on sharp inequalities for conjugate functions, where best constant inequalities were related to the existence of suitable subharmonic minorants; see [8, 10]. This method was later developed in a effective plurisubharmonic form by Hollenbeck and Verbitsky [11], and has been widely used to determine exact norms and sharp constants for Riesz projections and Hilbert transforms [1, 10, 16].

2. The proof of Theorem 1.1

The main difficulty in applying the subharmonic minorant method lies in constructing a suitable subharmonic function adapted to the desired integral inequality. In the following content, we put

$$p := 4\left(1 - \frac{1}{q}\right).$$

Let us begin by considering the weighted arithmetic-geometric mean inequality:

$$\lambda a^q + 1 - \lambda \geq a^{\lambda q}, \quad 0 \leq \lambda \leq 1.$$

Set $\lambda := p/q$. Then $0 < \lambda \le 1$ for $1 < q < \infty$, and hence

$$\lambda a^q + 1 - \lambda - a^p \ge 0. \tag{4}$$

Now define

$$\Phi(w,z) := \lambda |w + \overline{z}|^q + 1 - \lambda - |w|^p, \quad (w,z) \in \mathbb{C}^2.$$

Next, we turn to finding a suitable plurisubharmonic function that bounds Φ from below. It is clear that $\Phi \ge 0$ whenever $w = 0$. Observe that when $w \ne 0$, it holds that

$$\begin{aligned}\Phi(w,z) =& \lambda \frac{\big||w|^2 + \overline{zw}\big|^q}{|w|^q} + 1 - \lambda - |w|^p \\ \ge& \inf_{a>0} \left\{ \lambda \frac{|a^2 + zw|^q}{a^q} + 1 - \lambda - a^p \right\}.\end{aligned}$$

In particular, if we define

$$G(\zeta) := \inf_{a>0} \left\{ \lambda \frac{|a^2 + \zeta|^q}{a^q} + 1 - \lambda - a^p \right\}, \tag{5}$$

then

$$\Phi(z,w) \ge G(zw), \quad (z,w) \in \mathbb{C}^2. \tag{6}$$

The key point is that, although the infimum of a family of subharmonic functions need not be subharmonic in general, the function G defined here is indeed subharmonic.

Lemma 2.1. *The function G defined in* (5) *is subharmonic.*

Let us first prove Theorem 1.1 on the assumption that Lemma 2.1 holds, and then proceed to verify Lemma 2.1 in the next section.

Proof of Theorem 1.1. Suppose

$$f = g + \overline{h}$$

where g, h are analytic polynomials and $h(0) = 0$. Since G is subharmonic, the composition

$$z \mapsto G\big(g(z) \cdot h(z)\big)$$

is also subharmonic. By the sub-mean-value principle, we have

$$\begin{aligned}&\lambda \int_{\mathbb{T}} |g + \overline{h}|^q \, dm + 1 - \lambda - \int_{\mathbb{T}} |g|^p \, dm \\ =& \int_{\mathbb{T}} \Phi(g,h) \, dm \ge \int_{\mathbb{T}} G(g \cdot h) \, dm \\ \ge& G\big(g(0)h(0)\big) = G(0) = 0.\end{aligned}$$

It follows that

$$\int_{\mathbb{T}} |\mathcal{P} f|^p \, dm \le \lambda \int_{\mathbb{T}} |f|^q \, dm + 1 - \lambda.$$

This shows

$$\|\mathcal{P} f\|_{L^p} \le 1$$

provided that $\|f\|_{L^q} \le 1$. The proof of Theorem 1.1 is completed. □

3. Proof of Lemma 2.1

It remains to verify that the function G defined in (5) is indeed subharmonic. The case $q = p = 2$ is trivial, and in this case $G(\zeta) = 2\,\mathrm{Re}\,\zeta$. In the following content we always assume $q \neq 2$, and set

$$P := p/2, \qquad Q := q/2, \qquad \text{and recall that } \lambda = p/q = P/Q.$$

Then the relation $p = 4(1 - 1/q)$ gives

$$PQ = 2Q - 1 > 0, \qquad Q - P = \frac{(Q-1)^2}{Q} > 0. \tag{7}$$

A change of variable ($x = a^2$) gives

$$G(\zeta) = \inf_{x>0} F(x, \zeta),$$

where

$$F(x,\zeta) := \frac{P}{Q}\frac{|x+\zeta|^{2Q}}{x^Q} + 1 - \frac{P}{Q} - x^P, \qquad (x,\zeta) \in (0,\infty)\times\mathbb{C}. \tag{8}$$

Next, we compute the partial derivatives to locate the minimizer in x of $F(x,\zeta)$ for each fixed ζ. We adopt the notation

$$\partial_1 := \frac{\partial}{\partial x}, \quad \partial_2 = \frac{\partial}{\partial \zeta}, \quad \text{and } \overline{\partial}_2 := \frac{\partial}{\partial \overline{\zeta}}.$$

Lemma 3.1. *For each $\zeta \in \mathbb{C}$, there exists a unique*

$$x_{\min}(\zeta) \in (0,\infty)$$

such that

$$G(\zeta) = \inf_{x>0} F(x,\zeta) = F\big(x_{\min}(\zeta), \zeta\big).$$

Moreover, the minimizer function

$$x_{\min} : \zeta \mapsto x_{\min}(\zeta)$$

is smooth and $x_{\min}(\zeta) > |\zeta|$ for all $\zeta \in \mathbb{C}$.

Proof. Since $2Q > 1$, the function F is continuously differentiable throughout $(0,\infty)\times\mathbb{C}$. A standard calculation gives

$$\partial_1 F(x,\zeta) = \begin{cases} Px^{P-1}\Big(x^{-P-Q}|x+\zeta|^{2Q-2}(x^2-|\zeta|^2) - 1\Big), & \text{if } \zeta \neq 0 \text{ and } x \neq -\zeta, \\ -Px^{P-1}, & \text{if } \zeta \neq 0 \text{ and } x = -\zeta, \\ Px^{P-1}\big(x^{Q-P} - 1\big), & \text{if } \zeta = 0. \end{cases} \tag{9}$$

In the case $\zeta = 0$, since $Q > P$, we see

$$x_{\min}(0) = 1.$$

For fixed $\zeta \neq 0$, we observe that

$$\lim_{x\to 0} F(x,\zeta) = \lim_{x\to\infty} F(x,\zeta) = +\infty.$$

Thus the minimum is indeed attained, and every minimizer point is a solution of

$$\partial_1 F(x,\zeta) = 0. \tag{10}$$

It follows from (9) that every minimizing point satisfies $x > |\zeta|$.

Note that for $x > |\zeta|$, the stationary equation (10) is equivalent to

$$\frac{|x+\zeta|^{2Q-2}(x^2-|\zeta|^2)}{x^{P+Q}} = 1. \tag{11}$$

We claim that the left-hand side is strictly increasing in x on $(|\zeta|,\infty)$. Indeed,

$$\partial_1 \log\left(\frac{|x+\zeta|^{2Q-2}(x^2-|\zeta|^2)}{x^{P+Q}}\right) = \operatorname{Re}\frac{2(Q-1)}{x+\zeta} + \frac{2x}{x^2-|\zeta|^2} - \frac{P+Q}{x}.$$

Note that

$$\frac{1}{x+|\zeta|} \le \operatorname{Re}\frac{1}{x+\zeta} \le \frac{1}{x-|\zeta|}.$$

Substituting the left-hand estimate when $Q > 1$ and the right-hand estimate when $Q < 1$, we obtain, after putting the resulting terms over a common denominator,

$$\begin{aligned}
&\partial_1 \log\left(\frac{|x+\zeta|^{2Q-2}(x^2-|\zeta|^2)}{x^{P+Q}}\right) \\
&\quad\ge \frac{(Q-P)x^2 - 2|Q-1|\,x|\zeta| + (P+Q)|\zeta|^2}{x(x^2-|\zeta|^2)} \\
&\quad= \frac{\big(|Q-1|x - Q|\zeta|\big)^2 + (2Q-1)|\zeta|^2}{Qx(x^2-|\zeta|^2)} > 0.
\end{aligned}$$

The last equality uses the identities in (7), and the strict inequality follows from $x > |\zeta|$ and $2Q > 1$. Thus (10) has the unique solution

$$x = x_{\min}(\zeta) > |\zeta|.$$

Moreover, it is clear that F is smooth on the domain

$$\mathcal{R} := \Big\{(x,\zeta) \in (0,\infty)\times\mathbb{C} : x > |\zeta|\Big\}.$$

Since the left-hand side of (11) is strictly increasing with x, it follows that

$$\partial_1^2 F(x_{\min}(\zeta),\zeta) > 0, \quad \zeta \in \mathbb{C}.$$

Since $\big(x_{\min}(\zeta),\zeta\big) \in \mathcal{R}$ for every $\zeta \in \mathbb{C}$, an application of the implicit function theorem to the equation (10) shows that $x_{\min}$ is smooth. □

By Lemma 3.1, we conclude that

$$G(\zeta) = F(x_{\min}(\zeta), \zeta),$$

and hence G is also smooth. Thus it remains to show

$$\Delta G \geq 0.$$

We now calculate the Laplacian directly. To simplify the notation, we write

$$\partial := \frac{\partial}{\partial \zeta}, \qquad \overline{\partial} := \frac{\partial}{\partial \overline{\zeta}}.$$

Moreover, in the calculation below, x always denotes $x_{\min}(\zeta)$, and every partial derivative of F is evaluated at the point $(x_{\min}(\zeta), \zeta)$.

Proof of Lemma 2.1. Differentiating

$$\partial_1 F(x, \zeta) = 0 \tag{12}$$

with respect to ζ and $\overline{\zeta}$, respectively, and using $\partial_1^2 F(x, \zeta) > 0$, we obtain

$$\partial x_{\min}(\zeta) = -\frac{\partial_1 \partial_2 F(x, \zeta)}{\partial_1^2 F(x, \zeta)}, \qquad \bar{\partial} x_{\min}(\zeta) = -\frac{\partial_1 \overline{\partial}_2 F(x, \zeta)}{\partial_1^2 F(x, \zeta)}. \tag{13}$$

On the other hand, the chain rule and (12) give

$$\partial G(\zeta) = \partial_1 F(x, \zeta)\, \partial x_{\min}(\zeta) + \partial_2 F(x, \zeta) = \partial_2 F(x, \zeta).$$

Applying $\bar{\partial}$ to this identity and then using (13), we find that

$$\begin{aligned} \partial \bar{\partial} G(\zeta) &= \partial_1 \partial_2 F(x, \zeta)\, \bar{\partial} x_{\min}(\zeta) + \partial_2 \overline{\partial}_2 F(x, \zeta) \\ &= \partial_2 \overline{\partial}_2 F(x, \zeta) - \frac{\partial_1 \partial_2 F(x, \zeta)\, \partial_1 \overline{\partial}_2 F(x, \zeta)}{\partial_1^2 F(x, \zeta)} \\ &= \frac{\partial_1^2 F(x, \zeta)\, \partial_2 \overline{\partial}_2 F(x, \zeta) - |\partial_1 \partial_2 F(x, \zeta)|^2}{\partial_1^2 F(x, \zeta)}. \end{aligned} \tag{14}$$

Here we used the fact that F is real-valued, and hence $\partial_1 \overline{\partial}_2 F = \overline{\partial_1 \partial_2 F}$.

Now it suffices to show the numerator in the last expression is nonnegative. Applying ∂_1 to (9) and then substituting (11), we obtain

$$\partial_1^2 F(x, \zeta) = P x^{P-1} \left(\frac{2x}{x^2 - |\zeta|^2} + \operatorname{Re} \frac{2(Q-1)}{x + \zeta} - \frac{P + Q}{x} \right).$$

Similarly, applying ∂_2 to (9) and then substituting (11) gives

$$\partial_1 \partial_2 F(x, \zeta) = P x^{P-1} \left(\frac{Q-1}{x + \zeta} - \frac{\overline{\zeta}}{x^2 - |\zeta|^2} \right).$$

Finally, direct differentiation of (8) gives

$$\begin{aligned} \partial_2 F(x, \zeta) &= P x^{-Q} |x + \zeta|^{2Q-2} (x + \overline{\zeta}), \\ \partial_2 \overline{\partial}_2 F(x, \zeta) &= P Q x^{-Q} |x + \zeta|^{2Q-2} = \frac{P Q x^P}{x^2 - |\zeta|^2}, \end{aligned}$$

where the last equality again follows from (11).

Substituting these three formulas into (14) gives

$$\begin{aligned}
&\frac{\partial_1^2 F(x,\zeta)\,\partial_2\overline{\partial}_2 F(x,\zeta) - |\partial_1\partial_2 F(x,\zeta)|^2}{P^2 x^{2P-2}} \\
&\quad = \frac{Qx}{x^2-|\zeta|^2}\left(\frac{2x}{x^2-|\zeta|^2} + 2(Q-1)\operatorname{Re}\frac{1}{x+\zeta} - \frac{P+Q}{x}\right) \\
&\qquad - \left|\frac{Q-1}{x+\zeta} - \frac{\overline{\zeta}}{x^2-|\zeta|^2}\right|^2 \\
&\quad = \frac{2Qx^2-|\zeta|^2}{(x^2-|\zeta|^2)^2} - \frac{Q(P+Q)}{x^2-|\zeta|^2} \\
&\qquad + \underbrace{\frac{2Q(Q-1)x}{x^2-|\zeta|^2}\operatorname{Re}\frac{1}{x+\zeta} - \frac{(Q-1)^2}{|x+\zeta|^2} + \frac{2(Q-1)}{x^2-|\zeta|^2}\operatorname{Re}\frac{\zeta}{x+\zeta}}_{\text{denoted by } \mathcal{T}}.
\end{aligned}$$

Since

$$\operatorname{Re}\frac{1}{x+\zeta} = \frac{x+\operatorname{Re}\zeta}{|x+\zeta|^2}, \qquad \operatorname{Re}\frac{\zeta}{x+\zeta} = \frac{x\operatorname{Re}\zeta+|\zeta|^2}{|x+\zeta|^2},$$

the last three terms in the preceding display can be combined as

$$\begin{aligned}
\mathcal{T} &= \frac{Q-1}{(x^2-|\zeta|^2)|x+\zeta|^2}\Big(2Qx(x+\operatorname{Re}\zeta) - (Q-1)(x^2-|\zeta|^2) \\
&\qquad + 2(x\operatorname{Re}\zeta+|\zeta|^2)\Big) \\
&= \frac{Q-1}{(x^2-|\zeta|^2)|x+\zeta|^2}\Big((Q+1)x^2 + 2(Q+1)x\operatorname{Re}\zeta + (Q+1)|\zeta|^2\Big) \\
&= \frac{Q^2-1}{x^2-|\zeta|^2}.
\end{aligned}$$

As a consequence,

$$\begin{aligned}
&\frac{\partial_1^2 F(x,\zeta)\,\partial_2\overline{\partial}_2 F(x,\zeta) - |\partial_1\partial_2 F(x,\zeta)|^2}{P^2 x^{2P-2}} \\
&\quad = \frac{2Qx^2-|\zeta|^2}{(x^2-|\zeta|^2)^2} - \frac{Q(P+Q)}{x^2-|\zeta|^2} + \frac{Q^2-1}{x^2-|\zeta|^2} \\
&\quad = \frac{(2Q-1-PQ)x^2 + PQ|\zeta|^2}{(x^2-|\zeta|^2)^2} \\
&\quad = \frac{PQ|\zeta|^2}{(x^2-|\zeta|^2)^2} \geq 0,
\end{aligned}$$

where the last equality follows from $2Q-1-PQ=0$ in (7). Since $\partial_1^2 F(x,\zeta) > 0$, formula (14) now shows that

$$\partial\bar{\partial} G(\zeta) = \frac{P^2 x^{2P-2} PQ|\zeta|^2}{(x^2-|\zeta|^2)^2 \partial_1^2 F(x,\zeta)} \geq 0.$$

Thus G is subharmonic. The proof is completed. □

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